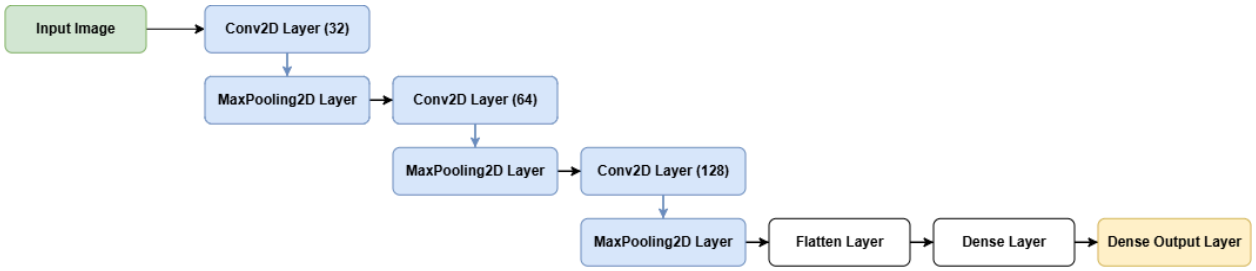
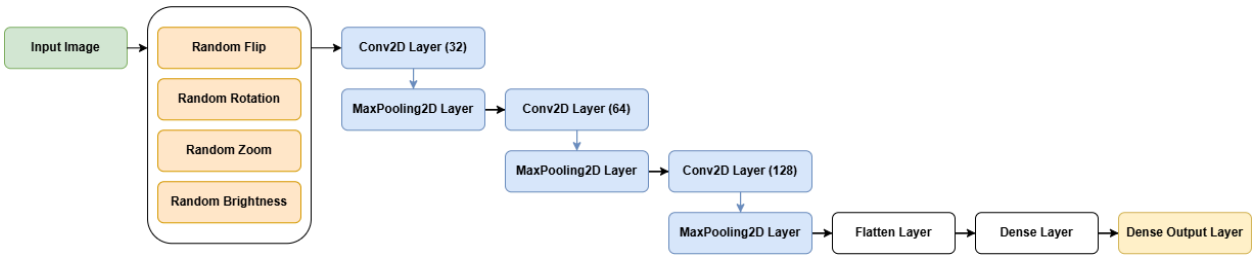


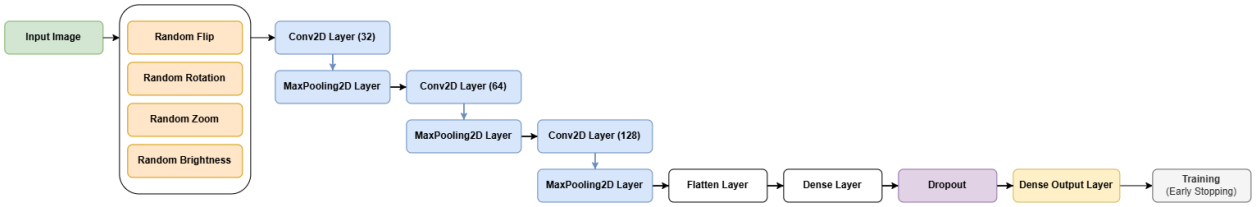
Model Architecture Diagram - Original CNN (Task 2)



Model Architecture Diagram - Modified CNN (Task 3)



Model Architecture Diagram - Applied Data Segmentation CNN (Task 4)



Model Architecture Diagram - Applied Regularization CNN (Task 5)

Training Settings:

- Epochs: 10 (changed to 20 for Applied Regularization CNN in Task 5)
- Optimizer: Adam
- Loss Function: Sparse Categorical Cross Entropy

Data Augmentation methods used:

- Random Flip: Helps the model recognize an object even when its position or orientation is reversed.
- Random Rotation: Helps the model less sensitive to small changes in the camera angle.
- Random Zoom: Helps the model recognize objects whether they appear closer or farther away.
- Random Brightness: Helps the model handle images taken under different lighting conditions.

Regularization methods used:

- Dropout: Randomly ignores 50% of the neurons in that layer. This helps reduce overfitting.
- Early Stopping: Monitor validation loss, stop if validation loss does not improve for 3 consecutive epochs and restore the model from the epoch with the best validation loss.

Original CNN (Task 2)	Modified CNN (Task 3)	Applied Data Augmentation CNN (Task 4)	Applied Regularization CNN (Task 5)
71.50%	78.53%	81.00%	82.90%

Accuracy Comparison between 4 CNN models